FROM TREES TO TECH · ACTIVITY TTT-01 · BIOLOGY + ENERGY

MudWatt Bioelectric League

Can your team make mud generate electricity and defend your design with data?

Win condition: Highest combined score from peak reading, daily data quality, controlled-variable design, redesign notes, and final explanation.

THE SCIENCE YOU ARE USING

Electrogenic bacteria. Some bacteria already in ordinary soil are electrogenic: as they digest nutrients, they push spare electrons onto surfaces outside their cells. Bury a graphite anode in oxygen-free mud and these microbes coat it, feeding it a steady stream of electrons.

Completing the circuit. Electrons leave the anode, run through the wire and the circuit board (lighting the LED on the way), and reach the cathode resting in the air on top. There they join oxygen and protons from the mud to form water. More food and more healthy microbes mean more electrons per second, so the LED blinks faster. Power is roughly voltage times current.

HOW TO BUILD AND RUN IT

01 **Inoculate the mud.** Gloves on. Pack the vessel with sponge-damp soil. The bacteria you need are already in the dirt, so no culture is required.

02 **Place the electrodes.** Bury the anode near the bottom and smooth mud over it so no air reaches it. Rest the cathode flat on top, exposed to the air.

03 **Add a measured fuel.** Mix in a small, measured amount of one approved fuel. Record exactly how much. This is the food that drives the blink rate.

04 **Connect and read a baseline.** Anode to (−), cathode to (+). Log the first blink interval or voltage. Day one is often weak; the biofilm needs time.

05 **Control one variable.** Pick ONE thing to test all week: fuel type, moisture, or anode contact. Change only that, predict the effect, and compare to baseline.

Safety: Gloves required. No food near mud station. Keep cells sealed except during setup. Wash hands after handling soil. Allergy: Do not use nut-containing food scraps. Preferred fuels are banana peel, sugar water, stale bread, or leaf litter.

MATERIALS

MudWatt or equivalent MFC kits	6
Fresh soil or mud	fill each vessel
Nitrile gloves	per student
Approved fuels	banana, sugar water, bread, leaf litter
Labels and tape	shared
Multimeters	optional

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): ______________________________

The ONE variable we are testing: ______________________________

Baseline result: ______________________________

After redesign: ______________________________

DEFEND YOUR DESIGN

What evidence made you change your design? ______________________________

If you ran it again, what is the ONE thing you would change? ______________________________

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Peak reading or blink performance	35
Daily data log quality	20
Controlled-variable design	15
Redesign or maintenance plan	15
Final explanation	15
Total	**100**